

JEE MAIN REVISION MASTER: STRAIGHT LINES AND PAIR OF STRAIGHT LINES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 12 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Distance, Slopes & Forms of Straight Line	8 PYQs	Straight Lines
Moderate	Pair of Straight Lines & Angle Bisectors	5 PYQs	Homogeneous Lines

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Distance, Slopes & Forms of Straight Line	$d = \frac{ a x_1 + b y_1 + c }{\sqrt{a^2 + b^2}}$, $d_{\text{parallel}} = \frac{ c_1 - c_2 }{\sqrt{a^2 + b^2}}$, $\text{Foot: } \frac{x - x_1}{a} = \frac{y - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}$
Pair of Straight Lines & Angle Bisectors	$\tan \theta = \left \frac{2\sqrt{h^2 - ab}}{a + b} \right $, $\text{Perpendicular } \iff a + b = 0$, $\text{Coincident } \iff h^2 = ab$, $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Distance, Slopes & Forms of Straight Line (High Yield — 8 PYQs)

Core Principle: Linear equations, distance formulas, family of lines $L_1 + \lambda L_2 = 0$, and coordinates of reflections (images) in lines.

■ **Caution / JEE Pitfall:** For distance between parallel lines, ensure coefficients of x and y are MADE IDENTICAL before subtracting $c_1 - c_2$.

■ **Speed Trick:** Image of point (x_1, y_1) in line $ax + by + c = 0$ is given by $(x - x_1 / a) = (y - y_1 / b) = -2(ax_1 + by_1 + c) / (a^2 + b^2)$.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "The system of equations $4x+6y=5$ and $8x+12y=10$ has..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "A line passes through $(-1,-3)$ and perpendicular to $x+6y=5$. Its x intercept is..." [Ans: N/A]
- [2024 Shift] Q#4 (MCQ): "The angle between the line $x+y=3$ and the line joining the points $(1,1)$ and $(-3,4)$ is..." [Ans: N/A]

3.2 Pair of Straight Lines & Angle Bisectors (Moderate Yield — 5 PYQs)

Core Principle: Joint equations of lines through origin, condition of perpendicularity ($a + b = 0$), and joint equation of angle bisectors.

■ **Caution / JEE Pitfall:** If $a + b = 0$, the lines represented by $ax^2 + 2hxy + by^2 = 0$ are mutually perpendicular regardless of value of h .

■ **Speed Trick:** General second degree equation represents pair of lines if $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The line L_2 passing through $(3, -1)$ divides the line segment L_1 joining the points $(-1, 2)$ and $(3, 6)$ in the ratio 1:3 intern..." **[Ans: N/A]**
- **[2023 Shift] Q#1 (MCQ):** "A line passes through $(2, 2)$ and is perpendicular to the line $3x + y = 3$. Its y -intercept is..." **[Ans: N/A]**
- **[2022 Shift] Q#2 (MCQ):** "If the straight line $2x - 3y + 17 = 0$ is perpendicular to the line passing through the points $(7, 17)$ and $(15, \beta)$, then β equals..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The line L_2 passing through $(3, -1)$ divides the line segment L_1 joining the points $(-1, 2)$ and $(3, 6)$ in the ratio 1:3 internally. The equation of line L_2 is:

- (A) $4x - 3y - 9 = 0$
- (B) $4x - 3y + 9 = 0$
- (C) $4x + 3y - 9 = 0$
- (D) $4x + 3y + 9 = 0$

Correct Answer: None

Solution Analysis: Let the points on line segment L_1 be $A(-1, 2)$ and $B(3, 6)$. The line L_2 divides this segment internally in the ratio 1:3. Using the section formula, the point of division P is $P\left(\frac{1 \cdot 3 + 3 \cdot (-1)}{1 + 3}, \frac{1 \cdot 6 + 3 \cdot 2}{1 + 3}\right) = P(0, 3)$. So, line L_2 passes through the points $(3, -1)$ and $(0, 3)$. Now find its slope: $m = \frac{3 - (-1)}{0 - 3} = \frac{4}{-3} = -\frac{4}{3}$. Using point-slope form with point $(0, 3)$: $y - 3 = -\frac{4}{3}(x - 0)$ $3y - 9 = -4x$ $4x + 3y - 9 = 0$. So, the equation of line L_2 is $4x + 3y - 9 = 0$. Correct option

Benchmark Problem 2: KCET 2025 (Q#2)

The system of equations $4x + 6y = 5$ and $8x + 12y = 10$ has

- (A) No solution
- (B) Infinitely many solutions
- (C) unique solution
- (D) Only two solutions

Correct Answer: None

Solution Analysis: Notice that the second equation is just twice the first: First equation: $4x + 6y = 5$. Multiply both sides by 2: $8x + 12y = 10$. Since both equations represent the same line, there are infinitely many solutions. Answer: Option B.